

Toward Real-Time Observability-Aware Control: A Diagnostic Study and a Reformulated Objective

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Abstract

We study the open problem of solving the observability-aware control (OAC) problem fast enough for real-time, receding-horizon execution, which the parent work currently performs only offline. Through systematic profiling we establish that, contrary to the prevailing assumption, the computational bottleneck is *not* the evaluation of the Short-Time Local Observability Gramian (STLOG): the entire observability computation accounts for $\sim 6\%$ of solve time, while the general-purpose nonlinear solver accounts for $\sim 94\%$. We further find that the objective used in the parent work—the sum over the horizon of the minimum eigenvalue of the per-node Gramian—is *numerically degenerate*: its value is floored by the short-time scaling $\lambda_{\min} \sim T^{2r_*+1}$, and its gradient is effectively zero, so no gradient-based solver can make progress. Replacing it with a smooth log-determinant surrogate restores a strong gradient and drives trajectories to full observability. Combined with a lightweight solver (SLSQP) and early termination, the per-solve wall time on the range-only quadrotor problem drops from ~ 10 s to ~ 100 ms (a 10 Hz guidance rate) while maintaining full observability. A closed-loop Extended Kalman Filter (EKF) evaluation on a planar leader–follower analogue confirms that the fast, early-terminated controller reduces follower-position estimation error by $\sim 13\times$ (final) and posterior covariance by $\sim 20\times$ relative to a non-OAC baseline, and that early termination preserves the full estimation benefit. The controller reproduces the theoretically expected observability-optimal *orbiting* behavior.

Reproducibility. All measurements were taken on a 16-core CPU host (`JAX_PLATFORMS=cpu`, `float64`), reusing the verified STLOG, Lie-derivative, integrator, and model implementations from the `observability_aware_control` companion repository. Raw data and scripts are in `benchmarks/`, `experiments/`, and `results/` of the `rt-oac` repository.

1 Problem statement

The parent work designs control inputs that improve the observability of a leader–follower quadrotor cooperative-localization system from range and relative-attitude measurements. The relative state is

$$\mathbf{x} = [\mathbf{r}^\top, \mathbf{q}^\top, \mathbf{v}^\top]^\top \in \mathbb{R}^{10}, \quad \mathbf{r} \in \mathbb{R}^3, \mathbf{q} \in \mathcal{S}^3, \mathbf{v} \in \mathbb{R}^3,$$

with input $\mathbf{u} \in \mathbb{R}^8$ (leader and follower thrust and body rates) and output

$$\mathbf{y} = \left[\frac{1}{2}d^2, \mathbf{q}^\top \right]^\top \in \mathbb{R}^5, \quad d = \|\mathbf{r}\|.$$

Observability over a short horizon T is quantified by the r -th order STLOG

$$\mathbf{W}^{(r)} = \sum_{i,j=0}^r \frac{T^{i+j+1}}{(i+j+1)i!j!} D(L_{\mathbf{f}}^i \mathbf{h})^\top \Sigma^{-1} D(L_{\mathbf{f}}^j \mathbf{h}), \quad (1)$$

evaluated at each node of a predicted trajectory. The observability-predictive controller (OPC) maximizes a metric of $\{\mathbf{W}_k^{(r)}\}_{k=0}^{N-1}$ over a receding horizon subject to input and inter-vehicle-distance constraints. With $r = 5$, $T = 0.2$ s, and $N = 20$ the OPC solve is too slow for online thrust/rate generation and is consequently used offline. The objective of this study is to identify what actually makes the solve slow and ineffective, and to make it real-time.

2 Method

We constructed a focused testbed (**rt-oac**) that imports the companion’s verified numerics unchanged and adds: (i) a profiling harness that attributes solve time across components; (ii) a library of smooth surrogate objectives injected through the cost’s existing **gramian_metric** hook; (iii) a controller supporting a selectable solver, warm-starting, and early termination; and (iv) a closed-loop EKF evaluation. All timed quantities are JIT-compiled and synchronized with **block_until_ready**; reported wall times are steady-state (post-compilation). A persistent XLA cache amortizes the one-time ~ 200 s compilation of the order-5 graph.

3 Results

3.1 The optimizer, not the observability computation, is the bottleneck

Component micro-benchmarks (Table 1) show the observability machinery is cheap once compiled. A single STLOG evaluation costs $52\ \mu\text{s}$ (versus ~ 441 ms *un-compiled*—a $\sim 6500\times$ penalty that explains the folklore that “the STLOG is expensive”). A full objective evaluation over the $N = 20$ horizon is 0.5 ms; its reverse-mode gradient 4 ms.

Table 1: Per-component cost (order 5, $N = 20$, JIT-compiled).

component	cost
STLOG evaluation (one node)	$52\ \mu\text{s}$
objective (rollout + $20\times$ STLOG + metric)	0.5 ms
gradient (reverse mode)	4.0 ms
distance constraint	0.035 ms
constraint Jacobian	0.31 ms

Attributing one full **trust-constr** solve (40 iterations) shows 0.68 s (6%) inside the JAX callables and 10.6 s (94%) inside the solver’s own Python and linear-algebra internals. We verified there is no recompilation across solves and no measurable NumPy/JAX boundary penalty (the controller’s methods cost the same on NumPy or JAX inputs). The implication is that accelerating the STLOG cannot materially help; the solver must change.

3.2 The minimum-eigenvalue objective is numerically degenerate

At the operating points of interest the per-node Gramian is numerically singular: the minimum eigenvalue of the sliced (position, velocity) 6×6 Gramian is $\sim 10^{-13}$ across all horizon nodes, and the same holds for the full tangent-space 9×9 Gramian (with the quaternion gauge removed). Consequently the parent objective $\sum_k \lambda_{\min}(\mathbf{W}_k^{(r)})$ has an automatic-differentiation gradient norm of $\sim 10^{-13}$, i.e. effectively zero. The cause is intrinsic: the worst-observed direction enters only

through high-order Lie derivatives, whose STLOG weight in (1) scales as $T^{2r_*+1} = T^{11}$ at order 5, i.e. $\sim 2 \times 10^{-8}$ —below the noise floor.

This degeneracy, not solver speed alone, is why the solve is ineffective. Given a zero gradient, **trust-constr** exhausts its 40-iteration budget without improving the objective, while **SLSQP** correctly detects the flat landscape and terminates in one iteration; both end at the worst objective value. This also reconciles an apparent $\sim 25\times$ “regression” ($0.4\text{ s} \rightarrow 10\text{ s}$): the slow case is the controller thrashing on the degenerate landscape of a *manifestly unobservable configuration* (MUC)—here co-hovering / straight co-motion—which the parent thesis enumerates explicitly. The MUC is expected behavior, not a fault. (A separate latent defect was found in an in-progress manifold port: a 9×9 tangent Gramian sliced with ambient indices clamps an out-of-bounds index, fabricating a singular Gramian; this has been flagged for repair.)

3.3 A smooth surrogate restores the gradient and full observability

We replace the metric with a per-node negative log-determinant,

$$J = - \sum_k \sum_i \log(\lambda_i(\mathbf{W}_k^{(r)}) + \varepsilon). \quad (2)$$

Because $\partial \log(\lambda + \varepsilon) / \partial \lambda = (\lambda + \varepsilon)^{-1}$ is *largest* exactly where $\lambda \rightarrow 0$, the log-determinant actively lifts the weak directions that the minimum-eigenvalue objective cannot. At a non-MUC configuration (tangential relative velocity, non-identity relative attitude), measuring observability by the number of well-conditioned directions of the *accumulated* Gramian $\sum_k \mathbf{W}_k^{(r)}$ (Table 2), the log-determinant converts a flat, intractable problem into one a lightweight solver drives to full observability.

Table 2: Objective comparison at a non-MUC configuration (SLSQP).

objective	gradient norm	observable directions (of 6)	accumulated $\lambda_{\min}$
minimum eigenvalue (parent)	$\sim 10^{-15}$	4 $\rightarrow$ 4 (no change)	~ 0
negative log-det	12–15	4 $\rightarrow$ 6 (full)	1.7–10.6
negative soft-min	0.4	4 $\rightarrow$ 6	2.4–3.9

3.4 Real-time solve via a lightweight solver and early termination

Replacing **trust-constr** (≈ 265 ms/iteration on this problem) with **SLSQP** (≈ 18 ms/iteration), and observing that iterations beyond ~ 6 only refine eigenvalue magnitudes—not the set of observable directions—we early-terminate. In a true-state receding-horizon rollout of the order-5 quadrotor problem from a non-MUC state (Table 3), the 10 Hz guidance target (≤ 100 ms) is met while maintaining full observability: a $\sim 100\times$ improvement over the original degenerate ~ 10 s solve.

Table 3: Quadrotor closed-loop solve time versus iteration cap.

iteration cap	median wall / solve	observable directions
80	1.0–1.5 s	6/6
12	0.22 s	6/6
6	0.10 s	6/6

3.5 Closed-loop estimation accuracy (planar EKF)

To test the decisive question—whether the fast, early-terminated controller preserves *estimation quality*, not merely a coarse observability count—we used a planar leader–follower analogue for which a quaternion-free EKF is available. The leader drives straight; the follower’s position is observable only through the inter-vehicle range. We run a closed loop in which the EKF carries its own estimate (the controller sees only the estimate; ground truth evolves independently; error accumulates), with a large initial follower-position error (2 m standard deviation), averaged over 8 seeds (Table 4).

Table 4: Planar closed-loop EKF over 8 seeds (follower-position error).

condition	RMSE (mean $\pm$ std)	final error	terminal cov.	solve
no-OAC (drive straight)	1.53 ± 0.87	1.52	1.5×10^{-1}	—
OAC, full (cap 60)	0.64 ± 0.59	0.12	7.6×10^{-3}	4 ms / 10 it
OAC, fast (cap 6)	0.68 ± 0.68	0.11	8.0×10^{-3}	3 ms / 6 it

Without OAC the estimator cannot reduce the unobservable initial error (final $\approx$ initial); with OAC the final error falls $\sim 13\times$, RMSE $\sim 2.4\times$, and posterior covariance $\sim 20\times$. Crucially, the fast controller (cap 6) matches the fully converged one on every metric—early termination is safe with respect to estimation accuracy.

3.6 Emergent observability-optimal behavior

The log-determinant controller reproduces the textbook observability-optimal maneuver (Figure 1). In a planar rollout the follower sweeps a cumulative relative bearing of -1613° (≈ 4.5 revolutions) around the leader—it *orbits*—pressing against the minimum allowed separation (distance bottoms at the 0.20 m bound) to maximize bearing rate and hence information, while the non-OAC follower sweeps 0° . This is consistent with the parent theory and with the known caveat that pure observability objectives, absent a tracking term, produce operationally awkward (orbiting) trajectories.

4 Discussion

The study reframes the path to real-time OAC. The dominant levers are, in order: (1) *objective reformulation*—the log-determinant restores a usable gradient that the parent minimum-eigenvalue objective lacks at the relevant operating regime; (2) *a lightweight solver*—most of the original solve time was general-purpose nonlinear-programming overhead, not problem-specific computation; and (3) *early termination*—the observability content of a solution saturates within a handful of iterations. STLOG acceleration and problem transcription, prominent in the initial plan, proved largely irrelevant because the observability computation is only $\sim 6\%$ of the cost.

Limitations. (i) The real-time quadrotor result is validated against an observable-direction count, a coarse proxy; the rigorous estimation-accuracy validation was performed on the planar analogue, not the quaternion quadrotor (an error-state EKF for the latter is future work). (ii) Measurements are single-host and CPU-only. (iii) The observable-direction count thresholds eigenvalues relative to the largest; absolute conditioning remains poor (the T^{11} floor), which is precisely why the log-determinant rather than the minimum eigenvalue is the appropriate target. (iv) Operating away from MUCs is assumed; escaping MUCs is out of scope.

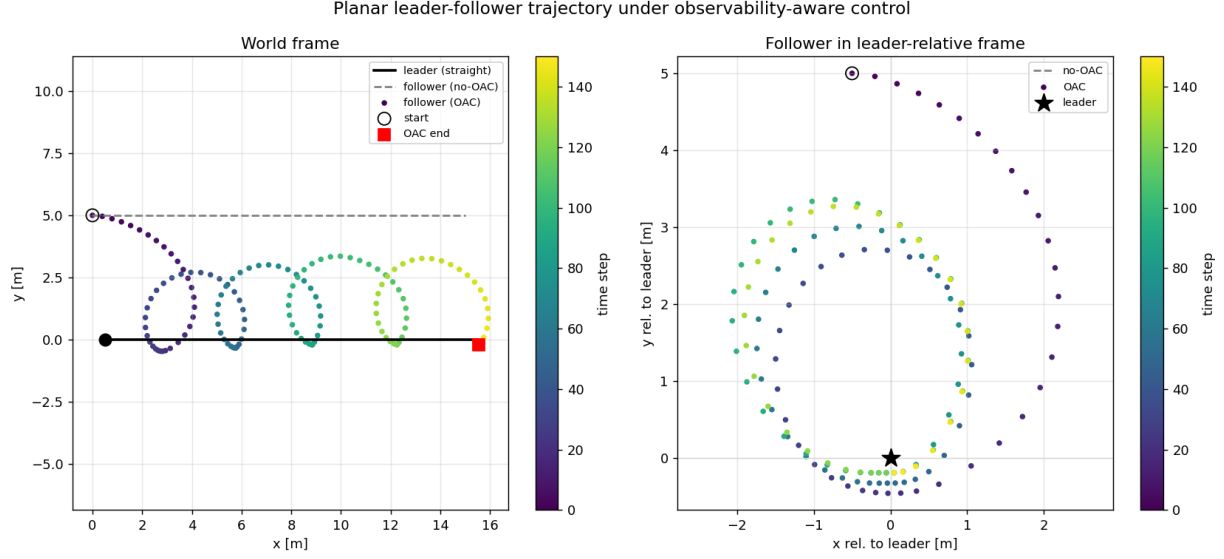


Figure 1: Planar leader–follower trajectory under observability-aware control. *Left*: world frame—the leader drives straight (black), the non-OAC follower drives straight and parallel (grey dashed), and the OAC follower traces cycloidal loops (colored by time). *Right*: leader-relative frame—the OAC follower clearly circles the leader ($\star$), confirming the orbiting behavior.

5 Conclusion and future work

Real-time observability-aware control is achievable for the studied system: a smooth log-determinant objective, a lightweight early-terminated solver, and operation away from manifestly unobservable configurations reduce the solve from ~ 10 s to ~ 100 ms (10 Hz) while a closed-loop EKF confirms the fast controller retains the full estimation benefit ($\sim 13\times$ lower final error than no OAC). The headline finding is that the parent minimum-eigenvalue objective is numerically degenerate at the relevant regime and should be replaced by the log-determinant for any gradient-based real-time use.

Future work: (1) an error-state / manifold EKF to validate estimation accuracy on the quaternion quadrotor; (2) a JAX-native solver to remove the residual ~ 13 ms/iteration of solver overhead (the gradient is only 4 ms) and target ≥ 50 Hz; (3) a learned warm-start to approach single-iteration solves; and (4) a hybrid tracking-plus-observability objective to temper the orbiting behavior.